

Sahoko Ishida

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Department of Computer Science, University of Oxford
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ACADEMIC POSITION

University of Oxford

Research Associate, Department of Computer Science

Oxford, UK

June 2024–present

Research topics: small area estimation, spatio-temporal modelling, survey sampling, food security, satellite imagery

A member of Machine Learning and Global Health Network

University of Bologna

Research Fellow, Department of Statistical Sciences

Bologna, Italy

January 2024–April 2024

EDUCATION

London School of Economics and Political Science (LSE),

PhD in Statistics. Advisor: Wicher Bergsma, Yunxiao Chen

London, UK

2019–2024

Thesis title: Statistical modelling with additive Gaussian process prior

London School of Economics and Political Science (LSE),

MSc in Statistics (Research), with Distinction

London, UK

2016–2017

Courses taken include: Statistical Computing, Machine Learning and Data Mining, GLM and Survival Analysis, Multilevel Modelling

Waseda University,

BA in Global Political Economy, Advisor: Shuhei Kurizaki

Tokyo, Japan

2012–2016

Focused on quantitative methods for political science and economics, minored in Data Analysis

RESEARCH INTERESTS

Gaussian process regression and kernel methods

Interpretable statistical learning, efficient algorithms for large-scale data, adaptive sampling, missing values

Spatio-temporal analysis

Small area estimation, forecasting and their application to food security, epidemiology, and environmental science

PRE-PRINT

1. **Ishida, S.**, & Bergsma, W. (2024). Efficient and Interpretable Additive Gaussian Process Regression on a Multidimensional Grid. *Under Review* Pre-print on <https://arxiv.org/abs/2305.07073>
2. Yang F., **Ishida, S.** Zhang, M. Jenson, D., Mishra S, Navott, J., & Flaxman S. (2024). Uncertainty-Aware Regression for Socio-Economic Estimation via Multi-View Remote Sensing. *In submission* Pre-print on <https://arxiv.org/abs/2411.14119>
3. **Ishida, S.**, Howes, A., Bradely, V., Semanova, E., Rashid, T., Dzumbunu, S., Bajaj, S., Singhal, G., Sejdinovic, D., Zvirere, H., Piovani, D., Passeri, S., Hachhethu, K., Husain, A., Kembo, G. D., Kairiza, T. & Flaxman S. (2025). Real-time small area estimation of food security in Zimbabwe: integrating mobile-phone and face-to-face surveys using joint multilevel regression and poststratification *Under Review* Pre-print on <https://arxiv.org/abs/2505.03517>

WORKING PAPERS

1. Riha, A., **Ishida, S.**, Howes, A., Flaxman, S., Semanova, E. Propensity-aware Bayesian optimisation with cost-cooling using Stan. Manuscript in preparation
2. **Ishida, S.**, et al. Adaptive learning sampling design: a new adaptive survey design paradigm. Manuscript in preparation
3. **Ishida, S.**, Osman, M., Cui, Z., Tang, K., & Flaxman, S. Bayesian small area estimation of micronutrient deficiency. Manuscript in preparation

SCHOLARSHIPS & AWARDS

LSE PhD Studentship	2019-2023
LSE Class Teacher Awards , LSE Eden Centre	2023
The Winton Prize for Best Dissertation , Department of Statistics, LSE	2017
Waseda University's 125th Anniversary Scholarship , School of Political Science and Economics	2012

CONFERENCE TALKS AND INVITED PRESENTATIONS

31st Conference of The International Environmetrics Society , Adelaide, Australia	December 2024
Royal Statistical Society International Conference 2024 , Brighton, UK	September 2024
International Conference on Spatio-Temporal Modelling , Lancaster, UK	July 2024
International Meeting of the Psychometric Society , Maryland, USA	July 2023
Spatial Statistics , Boulder, USA	July 2023
Nordic Conference in Mathematical Statistics , Gothenburg, Sweden	June 2023
Joint Statistical Seminars , Umeå University, Sweden	September 2022
Institute of Mathematical Statistics Annual Meeting , London, UK	June 2022
Spatial Statistics , Sitges, Spain	July 2019

TEACHING

Teacher , the African Institute for Mathematical Sciences	2025
<i>Teacher and lecturer on Bayesian inference and Gaussian process models in a short course: Modern Statistics and Machine Learning for Population Health in Africa</i>	
Graduate Teaching Assistant , the Department of Statistics, LSE	2019-2023
<i>Class Teacher and Coursework / Exam Marker for Master's level courses</i>	
ST411: Generalised Linear Modelling and Survival Analysis (R)	
ST425: Statistical Inference: Principles, Methods and Computation (R)	
ST451: Bayesian Machine Learning (Python, Stan)	
<i>Class Teacher and Coursework / Exam Marker for Bachelor's level courses:</i>	
ST102: Elementary Statistical Theory, ST104b: Statistics 2, ST107: Quantitative Methods	
ST115: Managing and Visualising Data (Python)	
ST201: Statistical Models and Data Analysis (R)	
ST308: Bayesian Inference (R, Stan, JAGS)	
<i>Award: LSE Class Teacher Awards (2023)</i>	
<i>Nomination: LSE Student Union Teaching Award for Excellent Feedback and Communication (2020)</i>	
Quantitative Advisor for Master's dissertation projects , LSE LIFE	2019-2022
<i>One-to-one help sessions for students across discipline, including Statistics, Economics, Psychology, Health Policy, Government, International Relations, Management</i>	

WORK EXPERIENCE

Department of Statistics, LSE Research Assistant for Dr. Francesca Panero working in collaboration with World Food Programme	London, UK 2023-2024
Grantham Research Institute on Climate Change and the Environment, LSE Research Assistant for Dr. Daire McCoy working on record linkage (address matching) problems	London, UK 2019
Eli Lilly Japan Statistical Analyst (full time) working in the team developing medication for type-II diabetes	Kobe, Japan 2018
School of Political Science and Economics, Waseda University Research Assistant for Associate Professor Shuhei Kurizaki on evaluation of Japan's defence policy	Tokyo, Japan 2016

PROFESSIONAL DEVELOPMENT AND SERVICE

Facilitator of bi-weekly meetings at the Machine Learning and Global Health network	2024-current
Reviewer for Statistical Methods & Applications	
Academic Representative, Department of Statistics, LSE	2021-2023
PhD/MSc Student Peer-to-Peer Mentor, Department of Statistics, LSE	2019-2021

OUTREACH

Speaker at Seien Girls' High School Career Day, Hamamatsu, Japan <i>Session organised by Soroptimist International Hamamamtsu</i>	2022
Private Educator, Tokyo, Japan <i>Tutoring and supporting pupils with special educational needs</i>	2016
Student Volunteer at ACTION (NPO), Olongapo, Philippines and Tokyo, Japan <i>Program targeting poverty alleviation for children on the street in Philippines</i>	2012